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Inventor(s)	Zhang; Shuo et al.

Staggered dual-side multi-chip interconnect

Abstract

Layout techniques for circuits on substrates are disclosed that address the multivariate problem of minimizing routing distances for high-speed I/O pins between circuits while simultaneously providing for the rapid provision of transient power demands to the circuits. The layout techniques may also enable improved thermal management for the circuits.

Inventors: Zhang; Shuo (Shenzhen, CN), Zhu; Eric (Shenzhen, CN), Zheng; Minto (Shenzhen, CN), Zhai; Michael (Chengdu, CN), Zhang; Town (Jiujiang, CN), Ma; Jie (Shenzhen, CN)

Applicant: NVIDIA Corp. (Santa Clara, CA)

Family ID: 86606446

Assignee: Nvidia Corp. (Santa Clara, CA)

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Primary Examiner: Ng; Sherman

Attorney, Agent or Firm: Carnelian Law

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority and benefit as a continuation of U.S. application Ser. No. 17/553,519, titled “Staggered Dual-Side Multi-Chip Interconnect”, filed on Dec. 16, 2021, the contents of which are incorporated herein by reference in their entirety.

BACKGROUND

(1) The insertion loss and crosstalk noise which are introduced by net routing path lengths on a printed circuit board (PCB) worsen as process nodes shrink (higher circuit density) and clock speeds increase. The path lengths for inter-chip communication in particular may limit performance.

(2) Conventionally, very large integrated circuit devices, such as graphics processing units (GPUs), central processing units (CPUs), and systems-on-a-chip (SoC) are mounted on a circuit board as depicted in FIG. 1 or FIG. 2. For example a central processing unit **102** and a graphics processing unit **104** may be mounted on the same side of a printed circuit board **106** as depicted in FIG. 1. In this case a minimum inter-chip path length is determined by the spacing between the sides of the chips (or chip packages, more generally). Routing lengths between certain performance-critical

pins on the chips may run several centimeters in some cases.

(3) Another approach is to mount a first chip package **202** on a first side **204** of a printed circuit board **206**, directly across from a second chip package **208** mounted on a second side **210** of the printed circuit board **206**. In this case the minimum inter-chip path length is determined by the thickness of the printed circuit board **206**. However the layout of other components utilized by either the first chip package **202**, second chip package **208**, or both becomes constrained. These other components are constrained to placement along side one chip package or the other.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced.

(2) FIG. **1** depicts an example of side-by-side chip package mounting on a printed circuit board.

(3) FIG. **2** depicts an example of stacked chip package mounting on a printed circuit board.

(4) FIG. **3A** depicts an example of stacked and staggered chip package mounting on a printed circuit board.

(5) FIG. **3B** depicts another example of stacked and staggered chip package mounting on a printed circuit board.

(6) FIG. **4A** depicts an example of transmit and receive pinout on a chip package in accordance with one embodiment.

(7) FIG. **4B** depicts an example of transmit and receive pinout on a chip package in accordance with another embodiment.

(8) FIG. **4C** depicts an example of transmit and receive pinout on a chip package in accordance with yet another embodiment.

(9) FIG. **5A** depicts yet another example of stacked and staggered chip package mounting on a printed circuit board.

(10) FIG. **5B** illustrates an aspect of the subject matter in accordance with one embodiment.

DETAILED DESCRIPTION

(11) Very large integrated circuit packages (e.g., GPUs, CPUs, SoCs) often draw large (relative to their average operating current) transient currents during certain operations, such as when initializing or performing certain power-intensive computational or input/output (I/O) operations. Components such as decoupling capacitors and power modules are ideally placed as close as possible to power pins of the chip packages to provide this transient energy quickly when needed. This competes with the requirements of high-speed I/O pins that communicate data and commands between chip packages.

(12) Layout techniques for chip packages on PCBs are disclosed that address the multivariate problem of minimizing routing distances for high-speed I/O pins between chip packages while simultaneously providing for the rapid provision of transient power demands to the chip packages, and/or thermal management for the chip packages.

(13) FIG. **1** depicts an example of side-by-side chip package mounting on a printed circuit board **106**. The printed circuit board **106** comprises a central processing unit **102** package mounted adjacent to a graphics processing unit **104** package, on the same side of the printed circuit board **106**. The physical constraints of such side-by-side mounting create a minimum inter-chip path length, which in turn imposes a maximum signaling rate between circuits on the central processing unit **102** and circuits on the graphics processing unit **104**. Although this example involves a central processing unit **102** and a graphics processing unit **104**, it is applicable generally to any types of computer chips.

(14) FIG. **2** depicts an example of stacked chip package mounting on a printed circuit board. A first

chip package **202** is mounted on a first side **204** of the printed circuit board **206**, and a second chip package **208** is mounted on a second side **210** of the printed circuit board **206**. Some or all of the IO pins of the first chip package **202** and second chip package **208** may be aligned with one another, minimizing the signaling length between them to be approximately the thickness of the printed circuit board **206**. However, this mounting layout comes with additional disadvantages in that other components on the printed circuit board **206** that communicate with or supply the first chip package **202** and second chip package **208** cannot be mounted across from them on the printed circuit board **206** because that space is now occupied by one of the chip packages.

(15) FIG. 3A depicts an example of stacked and staggered chip package mounting on a printed circuit board. The first chip package **202** is mounted on the first side **204** of the printed circuit board **206**, and the second chip package **208** is mounted on the second side **210** of the printed circuit board **206**, as before, but now staggered. The IO pins of the first chip package **202** and second chip package **208** that communicate with one another are specifically located in an interconnect region **302** of each of the packages, such that they are aligned with one another across the thickness dimension **304** of the printed circuit board **206** when the packages are mounted on opposite sides of the printed circuit board **206**. The interconnect region **302** may also be referred to herein as the “chip-to-chip critical inter-connection region”. The interconnect region **302** encompasses an area along one side of the first chip package **202** and also along one overlapping side of the second chip package **208**.

(16) The interconnect region **302** may provide direct routing connections (e.g., using vias) between transmit and receive pins of the first chip package and the second chip package.

(17) This mounting layout alleviates the prior disadvantages such that other components on the printed circuit board **206** that communicate with or supply the first chip package **202** (e.g., first chip package decoupling capacitor **306**) and second chip package **208** (e.g., second chip package decoupling capacitor **308**) can now be mounted across from them on the printed circuit board **206** because that space is no longer occupied by one of the chip packages. The first chip package decoupling capacitor **306** is located in the “mounting shadow” of the first chip package **202**, that is, within the periphery of the first chip package **202** but on the opposite side of the printed circuit board **206** from the first chip package **202**.

(18) FIG. 3B depicts an alternative in which the first chip package power module **310** and first chip package decoupling capacitor **306** are mounted adjacent to one another on the second side **210** of the printed circuit board **206**, across from the first chip package **202**, and the second chip package decoupling capacitor **308** and second chip package power module **312** are mounted adjacent to one another on the first side **204** of the printed circuit board **206**, across from the second chip package **208**.

(19) FIG. 4A depicts an example of transmit (TX) and receive (RX) pinout on a chip package in accordance with one embodiment. One or more uniform rows of transmit pins **402** are combined with one or more uniform rows of receive pins **404** along a single side of the package **406** for an integrated circuit die **408**. In some embodiments, all low-latency pins of the integrated circuit die **408**, or substantially all (e.g., $\geq 75\%$) that communicate with another particular integrated circuit are concentrated along a single side. A “low-latency pin” is a pin with an operating rate (e.g., clock frequency) that is the fastest or among the fastest (e.g., in the top 10-20%) for a chip, such that the length of its routes to other chips is a limiting factor for the performance of the chip or system overall. In other words, it is a performance-critical pin. A “uniform row” is one that comprises only pins of a particular function, such as only data transmit (output) pins or only data receive (input) pins. In one embodiment the receive pins **404** are arranged in pairs, as are the transmit pins **402**, resulting in two adjacent rows of transmit pins **402** adjacent to two adjacent rows of receive pins **404**. This is referred to herein as a “uniform row pair” configuration of pins.

(20) FIG. 4B depicts an example of transmit and receive pinout on a chip package in accordance with another embodiment. In this embodiment, the transmit pins **402** are 1:1 interleaved with the

receive pins **404** across each row, and each adjacent row is staggered, such that each pair of rows forms adjacent pairs of the transmit pins **402** and the receive pins **404** along its length. As with the embodiment in FIG. 4A, all or substantially all of the low-latency pins of the integrated circuit die **408** may be concentrated along a single side of the package **406**. This is referred to herein as a “interleaved staggered row pair” configuration of pins.

(21) FIG. 4C depicts an example of transmit and receive pinout on a chip package in accordance with yet another embodiment. Uniform rows of transmit pins **402** are interleaved with uniform rows of receive pins **404** along one side of the package **406**. This is referred to herein as a “interleaved uniform row” configuration of pins.

(22) The pinout of the side of the chip package to which the package **406** interfaces on the other side of the PCB is a mirror image of these pin layouts, with the transmit and receive pins swapped in position from those depicted in FIG. 4A-FIG. 4C. In other words, transmit pin/receive pin pairs are directly aligned with one another across the thickness dimension of the PCB.

(23) A layout of the type depicted in FIG. 3A, FIG. 3B, and FIG. 5A, where the chip packages are stacked and staggered on opposite sides of a PCB from one another, and the low latency/performance critical pins are primarily or completely located along a single side of each chip package, is referred to herein as a “stacked and staggered layout”.

(24) FIG. 5A and FIG. 5B depict other examples of a stacked and staggered chip package mounting on a printed circuit board, in which a first chip package heat sink **502** is mounted on the first chip package **202** and a second chip package heat sink **504** is mounted on the second chip package **208**. These layouts may prove efficient for managing temperatures of the chip packages especially during periods of higher power demand, while accommodating the routing constraints as previously described.

LISTING OF DRAWING ELEMENTS

(25) **102** central processing unit **104** graphics processing unit **106** printed circuit board **202** first chip package **204** first side **206** printed circuit board **208** second chip package **210** second side **302** interconnect region **304** thickness dimension **306** first chip package decoupling capacitor **308** second chip package decoupling capacitor **310** first chip package power module **312** second chip package power module **402** transmit pins **404** receive pins **406** package **408** integrated circuit die **502** first chip package heat sink **504** second chip package heat sink

(26) Various functional operations described herein may be implemented in logic that is referred to using a noun or noun phrase reflecting said operation or function. For example, an association operation may be carried out by an “associator” or “correlator”. Likewise, switching may be carried out by a “switch”, selection by a “selector”, and so on. “Logic” refers to machine memory circuits and non-transitory machine readable media comprising machine-executable instructions (software and firmware), and/or circuitry (hardware) which by way of its material and/or material-energy configuration comprises control and/or procedural signals, and/or settings and values (such as resistance, impedance, capacitance, inductance, current/voltage ratings, etc.), that may be applied to influence the operation of a device. Magnetic media, electronic circuits, electrical and optical memory (both volatile and nonvolatile), and firmware are examples of logic. Logic specifically excludes pure signals or software per se (however does not exclude machine memories comprising software and thereby forming configurations of matter).

(27) Within this disclosure, different entities (which may variously be referred to as “units,” “circuits,” other components, etc.) may be described or claimed as “configured” to perform one or more tasks or operations. This formulation—[entity] configured to [perform one or more tasks]—is used herein to refer to structure (i.e., something physical, such as an electronic circuit). More specifically, this formulation is used to indicate that this structure is arranged to perform the one or more tasks during operation. A structure can be said to be “configured to” perform some task even if the structure is not currently being operated. A “credit distribution circuit configured to distribute credits to a plurality of processor cores” is intended to cover, for example, an integrated circuit that

has circuitry that performs this function during operation, even if the integrated circuit in question is not currently being used (e.g., a power supply is not connected to it). Thus, an entity described or recited as “configured to” perform some task refers to something physical, such as a device, circuit, memory storing program instructions executable to implement the task, etc. This phrase is not used herein to refer to something intangible.

(28) The term “configured to” is not intended to mean “configurable to.” An unprogrammed FPGA, for example, would not be considered to be “configured to” perform some specific function, although it may be “configurable to” perform that function after programming.

(29) Reciting in the appended claims that a structure is “configured to” perform one or more tasks is expressly intended not to invoke 35 U.S.C. § 112(f) for that claim element. Accordingly, claims in this application that do not otherwise include the “means for” [performing a function] construct should not be interpreted under 35 U.S.C. § 112(f).

(30) As used herein, the term “based on” is used to describe one or more factors that affect a determination. This term does not foreclose the possibility that additional factors may affect the determination. That is, a determination may be solely based on specified factors or based on the specified factors as well as other, unspecified factors. Consider the phrase “determine A based on B.” This phrase specifies that B is a factor that is used to determine A or that affects the determination of A. This phrase does not foreclose that the determination of A may also be based on some other factor, such as C. This phrase is also intended to cover an embodiment in which A is determined based solely on B. As used herein, the phrase “based on” is synonymous with the phrase “based at least in part on.”

(31) As used herein, the phrase “in response to” describes one or more factors that trigger an effect. This phrase does not foreclose the possibility that additional factors may affect or otherwise trigger the effect. That is, an effect may be solely in response to those factors, or may be in response to the specified factors as well as other, unspecified factors. Consider the phrase “perform A in response to B.” This phrase specifies that B is a factor that triggers the performance of A. This phrase does not foreclose that performing A may also be in response to some other factor, such as C. This phrase is also intended to cover an embodiment in which A is performed solely in response to B.

(32) As used herein, the terms “first,” “second,” etc. are used as labels for nouns that they precede, and do not imply any type of ordering (e.g., spatial, temporal, logical, etc.), unless stated otherwise. For example, in a register file having eight registers, the terms “first register” and “second register” can be used to refer to any two of the eight registers, and not, for example, just logical registers 0 and 1.

(33) When used in the claims, the term “or” is used as an inclusive or and not as an exclusive or. For example, the phrase “at least one of x, y, or z” means any one of x, y, and z, as well as any combination thereof.

(34) As used herein, a recitation of “and/or” with respect to two or more elements should be interpreted to mean only one element, or a combination of elements. For example, “element A, element B, and/or element C” may include only element A, only element B, only element C, element A and element B, element A and element C, element B and element C, or elements A, B, and C. In addition, “at least one of element A or element B” may include at least one of element A, at least one of element B, or at least one of element A and at least one of element B. Further, “at least one of element A and element B” may include at least one of element A, at least one of element B, or at least one of element A and at least one of element B.

(35) The subject matter of the present disclosure is described with specificity herein to meet statutory requirements. However, the description itself is not intended to limit the scope of this disclosure. Rather, the inventors have contemplated that the claimed subject matter might also be embodied in other ways, to include different steps or combinations of steps similar to the ones described in this document, in conjunction with other present or future technologies. Moreover, although the terms “step” and/or “block” may be used herein to connote different elements of

methods employed, the terms should not be interpreted as implying any particular order among or between various steps herein disclosed unless and except when the order of individual steps is explicitly described.

(36) Having thus described illustrative embodiments in detail, it will be apparent that modifications and variations are possible without departing from the scope of the invention as claimed. The scope of inventive subject matter is not limited to the depicted embodiments but is rather set forth in the following Claims.

Claims

1. A device comprising: a first circuit package mounted on a first side of a substrate; and a second circuit package mounted on a second side of the substrate in a stacked and staggered layout with the first circuit package; vias forming a direct vertical connection through the substrate between input/output (IO) pins of the first circuit package and IO pins of the second circuit package in a region of the substrate overlapped by the first circuit package and the second circuit package; and wherein the IO pins of the first circuit package are located exclusively in rows extending end-to-end along a single side of the first circuit package, and the IO pins of the second circuit package are located exclusively in rows extending end-to-end along a single side of the second circuit package.
2. The device of claim 1, further comprising: a decoupling capacitor for the first circuit package mounted in a mounting shadow of the first circuit package.
3. The device of claim 2, further comprising: a heat sink for the first circuit package mounted in the mounting shadow of the first circuit package.
4. The device of claim 2, further comprising: a power module for the first circuit package mounted adjacent to the first circuit package on the first side of the substrate.
5. The device of claim 2, further comprising: a decoupling capacitor for the second circuit package mounted in a mounting shadow of the second circuit package.
6. The device of claim 5, further comprising: a heat sink for the second circuit package mounted in the mounting shadow of the second circuit package.
7. The device of claim 5, further comprising: a power module for the second circuit package mounted adjacent to the second circuit package on the second side of the substrate.
8. A device comprising: a first circuit and a second circuit mounted on opposite sides of a substrate in a stacked and staggered layout; and wherein peripheral regions of the first circuit and the second circuit that together form the entirety of an overlapped interconnect region of the stacked and staggered layout comprise one of a uniform row pair configuration of pins in direct vertical alignment, an interleaved staggered row pair configuration of pins in direct vertical alignment, and an interleaved uniform row configuration of pins in direct vertical alignment.
9. The device of claim 8, further comprising: a power module for the first circuit mounted adjacent to the first circuit on a first side of the substrate.
10. The device of claim 8, further comprising: a power module for the second circuit mounted adjacent to the circuit on a second side of the substrate.
11. The device of claim 8, wherein the first circuit comprises a graphics processing unit and the second circuit comprises a central processing unit.
12. The device of claim 8, wherein at least one of the first circuit and the second circuit comprises a system-on-a-chip.
13. A circuit board comprising: a first integrated circuit mounted on a first side of the circuit board; a second integrated circuit mounted on a second side of the circuit board in a stacked and staggered layout with the first integrated circuit; vertical vias configured in the circuit board in an overlapped interconnect region of the stacked and staggered layout, the vertical vias forming directly vertical connections between data transmitter pins of the first integrated circuit and data receiver pins of the second integrated circuit; and wherein the transmitter pins are located exclusively in rows

extending end-to-end along a single side of the first integrated circuit, and the receiver pins are located exclusively in rows extending end-to-end along a single side of the second integrated circuit.

14. The circuit board of claim 13, wherein a pinout of at least one of the integrated circuits in the overlapped interconnect region comprises a uniform row pair configuration of pins.

15. The circuit board of claim 13, wherein a pinout of at least one of the integrated circuits in the overlapped interconnect region comprises an interleaved staggered row pair configuration of pins.

16. The circuit board of claim 13, wherein a pinout of at least one of the integrated circuits in the overlapped interconnect region comprises an interleaved uniform row configuration of pins.

17. The circuit board of claim 13, further comprising: a decoupling capacitor for the first integrated circuit mounted on the circuit board in a mounting shadow of the first integrated circuit.

18. The circuit board of claim 17, further comprising: a heat sink for the first integrated circuit mounted on the circuit board in the mounting shadow of the first integrated circuit.

19. The circuit board of claim 17, further comprising: a decoupling capacitor for the second integrated circuit mounted on the circuit board in a mounting shadow of the second integrated circuit.

20. The circuit board of claim 19, further comprising: a heat sink for the second integrated circuit mounted on the circuit board in the mounting shadow of the second integrated circuit.

21. An electronic assembly comprising: a substrate comprising: a first side, a second side opposite the first side, and a plurality of vias extending directly vertically through the substrate from the first side to the second side; a first circuit package mounted on the first side of the substrate and including a first plurality of input/output (IO) connection points organized into rows extending end-to-end exclusively along a single side of the first circuit package; and a second circuit package mounted on the second side of the substrate in a stacked and staggered layout with the first circuit package and including a second plurality of IO connection points organized into rows extending end-to-end exclusively along a single side of the second circuit package; and the vias connecting the first plurality of IO connection points and the second plurality of IO connection points in an overlap region of the substrate overlapped by both the first circuit package and the second circuit package, the overlap region formed entirely by an area comprising the first plurality of IO connection points and the second plurality of IO connection points.

22. The electronic assembly of claim 21, wherein the first plurality of IO connection points comprise a plurality of pins.

23. The electronic assembly of claim 21, wherein the first plurality of IO connection points comprise a plurality of solder bumps.
